

Test Specifications: Payments Domain

Gema Event Management Platform
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Source files: `services/payment.service.ts`, `services/refund.service.ts`

PaymentService — getPaymentRouting

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-PAY-001	Happy path: platform Stripe routing for vendor without custom Stripe	Vendor exists in DB with <code>paymentMode != CUSTOM_STRIPE</code>	<code>vendorOrTeacherId</code> = valid vendor ID, <code>amount</code> = 100	<code>usesVendorStripe</code> : false, <code>stripeInstance</code> = platform stripe, <code>serviceFee</code> = vendor's effective commission rate applied to 100	High
TC-PAY-002	Happy path: vendor Stripe routing (custom secret key) with active subscription	Vendor has <code>paymentMode = CUSTOM_STRIPE</code> , active subscription, <code>stripeSecretKey</code> set	<code>vendorOrTeacherId</code> = vendor ID, <code>amount</code> = 500	<code>usesVendorStripe</code> : true, <code>platformCommission</code> : 0, <code>serviceFee</code> : 0, <code>vendorPayout</code> : 500	High
TC-PAY-003	Happy path: vendor Stripe routing via Connect (no secret key)	Vendor has <code>paymentMode = CUSTOM_STRIPE</code> , active subscription, <code>stripeConnectOnboardingComplete</code> : true, <code>stripeConnectAccountId</code> set	<code>vendorOrTeacherId</code> = vendor ID, <code>amount</code> = 200	<code>usesVendorStripe</code> : true, <code>vendorStripeAccountId</code> = connect account ID, <code>platformCommission</code> : 0	High
TC-PAY-004	Teacher fallback: vendor lookup misses, teacher found	No vendor with given ID, teacher exists with same ID	<code>vendorOrTeacherId</code> = valid teacher ID, <code>amount</code> = 150	Routing resolved using teacher's payment settings; no error thrown	Medium
TC-PAY-005	Commission calculation precision — files arithmetic	Vendor on platform Stripe with 5% commission	<code>amount</code> = 1.01	<code>serviceFee</code> = 0.05 (rounded via files), <code>vendorPayout</code> = 0.96	High
TC-PAY-006	Commission calculation — zero amount	Vendor on platform Stripe	<code>amount</code> = 0	<code>serviceFee</code> : 0, <code>vendorPayout</code> : 0	Medium
TC-PAY-007	Custom Stripe but subscription inactive	Vendor has <code>paymentMode = CUSTOM_STRIPE</code> but subscription is expired	<code>vendorOrTeacherId</code> = vendor ID, <code>amount</code> = 300	Falls back to platform Stripe with commission applied	High
TC-PAY-008	Custom Stripe but no keys configured	Vendor has <code>paymentMode = CUSTOM_STRIPE</code> , active subscription, but neither <code>stripeSecretKey</code> nor Stripe Connect fields set	<code>vendorOrTeacherId</code> = vendor ID, <code>amount</code> = 300	Falls back to platform Stripe with commission applied	High
TC-PAY-009	Vendor and teacher both not found	No vendor or teacher with given ID in DB	<code>vendorOrTeacherId</code> = non-existent ObjectId, <code>amount</code> = 100	Error thrown: "Vendor or Teacher not found"; fallback applies: <code>serviceFee</code> = 5% of 100 = 5.00	High
TC-PAY-010	DB error during vendor lookup	MongoDB connection failure	<code>vendorOrTeacherId</code> = valid ID, <code>amount</code> = 100	Catch block executes; fallback 5% commission applied; <code>usesVendorStripe</code> : false	High

PaymentService — createPaymentIntent

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-PAY-011	Happy path: create intent without vendor	Stripe configured	<code>amount</code> = 100, <code>currency</code> = "AED", <code>orderId</code> = valid ObjectId string, no <code>paymentIntentId</code> , <code>vendorId</code>	Returns { <code>clientSecret</code> , <code>paymentIntentId</code> }; <code>currency</code> forced to "aed"	High
TC-PAY-012	Happy path: create intent with vendor (platform routing)	Vendor exists on platform Stripe	<code>amount</code> = 250, <code>currency</code> = "AED", <code>orderId</code> , <code>vendorId</code>	Intent created on platform stripe; <code>serviceFee</code> in metadata matches routing info	High

TC-PAY-013	Happy path: create intent with vendor Stripe	Vendor uses custom Stripe key	amount = 500, vendorId	Intent created using vendor's Stripe instance	High
TC-PAY-014	Currency normalization — input non-AED currency	N/A	amount = 100, currency = "USD", orderId	Intent created with currency = "aed" regardless of input	Medium
TC-PAY-015	Display currency/amount passed through metadata	N/A	displayCurrency = "USD", displayAmount = 27.23	Metadata logged with display values; Stripe intent still uses AED	Low
TC-PAY-016	Customer ID attached when provided	Stripe customer exists	customerId = "cus_xxx"	customer field set on PaymentIntent params	Medium
TC-PAY-017	Idempotency key uniqueness — same orderId called twice rapidly	N/A	Same orderId, two rapid calls	Each call generates unique idempotency key (includes Date.now()); Stripe deduplicates if same key reused	Medium
TC-PAY-018	Stripe API failure during intent creation	Stripe returns 500 error	Any valid params	Error thrown: "Failed to create payment intent"	High
TC-PAY-019	Invalid vendorId causes routing fallback	Non-existent vendorId	vendorId = "000000000000000000000000"	Routing falls back to 5% commission; intent still created on platform Stripe	High
TC-PAY-020	Metadata merge with custom metadata	N/A	metadata = { "ref": "booking-123" }	Final PaymentIntent metadata contains both default fields and ref: "booking-123"	Low

RefundService — calculateRefundableAmount

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-PAY-021	Happy path: paid order with service fee and tax	N/A	order.paymentStatus = "paid", subtotal = 100, serviceFee = 5, tax = 5, couponDiscount = 0	refundAmount: 100, nonRefundableAmount: 10, serviceFee: 5, tax: 5	High
TC-PAY-022	Coupon discount reduces refundable amount	N/A	subtotal = 100, couponDiscount = 20, serviceFee = 4, tax = 4	refundAmount: 80, nonRefundableAmount: 8	High
TC-PAY-023	Order not paid — returns all zeros	N/A	order.paymentStatus = "pending"	refundAmount: 0, nonRefundableAmount: 0, serviceFee: 0, tax: 0	High
TC-PAY-024	Coupon discount larger than subtotal — no negative refund	N/A	subtotal = 10, couponDiscount = 15, serviceFee = 1, tax = 1	refundAmount: 0 (Math.max(0, -5) = 0)	High
TC-PAY-025	Zero service fee and tax	N/A	subtotal = 50, serviceFee = 0, tax = 0, paymentStatus = "paid"	refundAmount: 50, nonRefundableAmount: 0	Medium
TC-PAY-026	Missing serviceFee/tax fields default to 0	Order created without fee fields	subtotal = 60, serviceFee undefined, tax undefined	Treated as 0; refundAmount: 60	Medium

RefundService — processRefund

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-PAY-027	Happy path: user-requested refund within 24-hour window	Order paid, not refunded, within cancellation window, has paymentIntentId	orderId, reason, initiatedBy, cancellationType = "user_requested"	Stripe refund created; order status = "refunded"; seats returned to event; CancellationLog created	High
TC-PAY-028	Happy path: admin-cancelled refund ignores 24-hour rule	Order paid, past 24-hour window	cancellationType = "admin_cancelled"	Refund processed regardless of cancellation window	High
TC-PAY-029	Happy path: event-cancelled batch refund	Order paid	cancellationType = "event_cancelled"	Refund processed; order marked refunded; seats returned	High
TC-	Order already			Error thrown: "Order has already been	

PAY-refunded 030	order.status = "refunded"	Valid orderId	refunded"; transaction rolled back	High
TC- PAY- Order not paid 031	order.paymentStatus = "pending"	Valid orderId	Error thrown: "Order has not been paid"; transaction rolled back	High
TC- User requests PAY-cancellation outside 032 allowed window	canBeCancelledByCustomer() cancellationType = returns { canCancel: false } "user_requested"		Error thrown with reason from canBeCancelledByCustomer(); transaction rolled back	High
TC- PAY- Order not found 033	No order with given ID	orderId = non-existent ID	Error thrown: "Order not found"; transaction rolled back	High
TC- Stripe refund API PAY- failure 034	Stripe returns error during refund	Valid paid order	refundStatus set to "failed" on order; transaction rolled back; error propagated	High
TC- Free order / no PAY- paymentIntentId 035	Order paid but paymentIntentId is null	Valid paid order without intent	Order status set to "refunded" without Stripe call; seats returned	Medium
TC- Unlimited seats event PAY- skip seat 036 restoration	schedule.unlimitedSeats = true	Valid refund	Refund completes; availableSeats not modified on event schedule	Medium
TC- Event or schedule PAY- not found during seat 037 restoration	Event deleted after order placed	Valid refund	Refund still completes; seat restoration skipped gracefully	Medium
TC- Zero refundable PAY- amount 038	subtotal = 0 or fully discounted by coupon	Valid paid order	Error thrown: "No refundable amount"	High
TC- Transaction atomicity PAY- DB failure mid- 039 refund	Simulated DB failure after Stripe refund succeeds	Valid paid order	Transaction rolled back; order not marked refunded; requires reconciliation	High
TC- CancellationLog PAY- created with correct 040 metadata	Happy path refund	Valid paid order with customer and event	CancellationLog contains orderNumber, customerEmail, eventTitle, correct amounts	Medium